

Challenge (Habanero)

Q1. What is the standard form of a quadratic function?

- (A) $ax^2 + bx + c$
- (B) $ax + bx^2 + c$
- (C) $ax^2 - bx + c$
- (D) $ax + b$
- (E) $ax^3 + bx^2 + c$

Q2. Which of the following is a quadratic function in standard form?

- (A) $2x^2 - 3x + 1$
- (B) $x^2 + 2x - 3$
- (C) $x + 2$
- (D) $x^3 - 2x^2 + x$
- (E) $x^2 - 2x - 3x^2$

Q3. Identify the coefficient of the x^2 term in $3x^2 + 2x - 1$

- (A) 1
- (B) 2
- (C) 3
- (D) -1
- (E) -3

Q4. In the game 'Epic Quest', the score S is given by $S = -2x^2 + 12x + 50$, where x is the level number. What is the standard form of this quadratic function?

- (A) $-2x^2 + 12x + 50$
- (B) $2x^2 - 12x - 50$
- (C) $x^2 + 6x + 25$
- (D) $x^2 - 6x - 25$
- (E) $-x^2 - 6x + 25$

Q5. The data usage D of a gamer is given by $D = 0.5x^2 + 2x + 10$,

where x is the number of hours played. Identify the vertex of this parabola.

- (A) $(-2, 6)$
- (B) $(-1, 9)$
- (C) $(-4, 5)$
- (D) $(-2, 9)$
- (E) $(-1, 12)$

Q6. The coding logic for a game uses the quadratic function $f(x) = x^2 - 4x + 4$. What is the vertex of this parabola?

- (A) $(-2, 0)$
- (B) $(-1, 1)$
- (C) $(-4, 4)$
- (D) $(2, 0)$
- (E) $(0, 4)$

Q7. Which quadratic function has its vertex at $(1, -3)$?

- (A) $f(x) = (x - 1)^2 - 3$
- (B) $f(x) = -(x - 1)^2 - 3$
- (C) $f(x) = x^2 + 2x - 3$
- (D) $f(x) = x^2 - 2x - 3$
- (E) $f(x) = -x^2 + 2x + 3$

Q8. The score S of a player in a game is given by $S = -x^2 + 4x + 5$. What is the key feature of this quadratic function?

- (A) It opens upward
- (B) It opens downward
- (C) It has a maximum value
- (D) It has a minimum value
- (E) Both B and C

Open Ended Questions (FRQ)

Q9. A gamer's data usage is modeled by the quadratic function $D = 0.2x^2 + x + 5$, where x is the number of hours played. Determine the equation of the axis of symmetry and the vertex of this parabola. Show your work and explain your reasoning.

Q10. The coding logic for a game uses the quadratic function $f(x) = -2x^2 + 4x + 1$. Determine the key features of this quadratic function, including its vertex, axis of symmetry, and whether it opens upward or downward. Use this information to sketch the graph of the parabola.

Chef's Tips

- Tip 1: Ensure students understand the definition of quadratic functions and standard form.
- Tip 2: Emphasize the importance of identifying key features of quadratic functions, such as the vertex and axis of symmetry.
- Tip 3: Encourage students to use real-world examples, such as tech and gaming, to illustrate their understanding of quadratic functions.